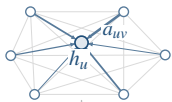


(a) Train

Graph attention



$$\min_{\theta} \sum_{(x,q) \in \mathcal{D}_k} \ell(\hat{q}_{\theta}(x), q)$$

θ_k

(b) Embed

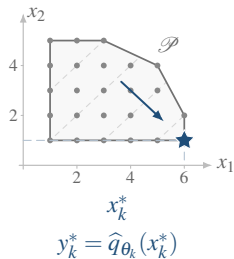
$$\min_{x,y} f(x,y)$$

$$x \in \mathcal{X}$$

$$g(x,y) \leq 0$$

$$y = \hat{q}_{\theta_k}(x)$$

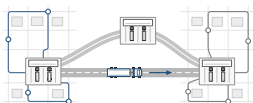
(c) Optimize



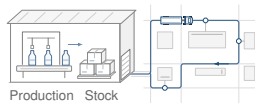
$$\mathcal{D}_{k+1} = \mathcal{D}_k \cup \{(x_k^*, q_k^{\text{obs}})\}$$

Evaluate / observe

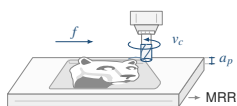
Transfer-hub networks



Production and distribution



Machining operations



Oil and gas drilling

